

Year 7 • Lesson 10.07 Analysing dot plots and stem-and-leaf plots: answers (teacher copy)

10.07 Analysing dot plots and stem-and-leaf plots (booklet p. 3)

Warm-up

- 4
- 4
- 13
- 18

WE DO examples

- Ex 1: range 5, mode 1
- Ex 2: median 1, mean 1.6
- Ex 3: range 4, median 4.5, mode 6, mean 4.25
- Ex 4: range 27, median 29, mode 34, mean 28.33
- Ex 5: with 15: mean 3.45, median 2; without: mean 2.3, median 2
- Ex 6: the median (or mode), because the outlier 15 pulls the mean up

Set A

- 15
- 6
- 6
- 2
- 2
- 32
- 2.13
- 6
- 5
- 2/15

Set B

- 10
- 24
- 53
- 29
- 31
- 36.5
- 367
- 36.7
- 2
- 35 years

Set C

- 9
- 2
- 2.5
- 2.9
- 9; it makes the mean bigger
- 35 min
- 25 min
- 25 min
- 24.86 min
- 3

Set D

- range 5, median 8, mode 8, mean 7.65
- mean 7.38, median 8
- The mean: the low score of 2 lowers the total, but the middle position barely moves
- No: total is 153; for a mean of 8 with 21 students the total must be 168, so they would need 15 out of 10
- range 53, median 21, mode 21, mean 24.47
- 68: it is far from all the other ages (15 to 34)
- mean 21.36, range 19
- The median (or mode), because the outlier pulls the mean up
- e.g. 2, 4, 4, 4, 5, 6, 7, 8, 8 drawn as a dot plot (mode 4, median 5, range 6)
- Values are balanced on both sides of the middle, so the mean sits in the middle too

Extension

- Total $5 \times 28 = 140$; $140 - 112 = 28$, so the leaf is 8
- Any value of 6 or more
- e.g. 1, 2, 3, 4, 5, 30: one large value pulls the mean up

Exit ticket

- Level 1: 3
- Level 2: median 3, range 7
- Level 3: mean 2.91; without the 8 it drops to 2.4